

SAVANNA AGCM, AB, Canada

WMO: 715470

Lat: 56.0761N

Lon: 119.3447W

Elev: 2306

StdP: 13.51

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.5	-18.9	-29.9	1.1	-23.4	-24.9	1.5	-18.6	29.5	35.6	27.0	34.5	4.3	270	0.705

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	80.5	59.9	77.3	58.7	74.4	57.5	62.9	74.5	61.2	72.9	59.5	70.8	9.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.7	80.3	67.6	56.7	74.7	64.5	54.9	69.9	63.3	29.6	74.4	28.3	73.1	27.1	71.1	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	21.0	18.6	-29.4	85.5	4.8	3.2	-32.9	87.8	-35.7	89.7	-38.4	91.4	-41.8	93.7
			-29.7	66.1	4.7	3.4	-33.1	68.5	-35.8	70.6	-38.5	72.5	-41.9	75.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	13.5	12.8	21.4	37.0	50.6	57.2	61.0	59.3	50.6	37.7	20.1	10.7	(6)
(7)		DBStd	21.74	16.66	15.24	14.12	9.00	7.92	5.60	4.80	6.05	8.20	9.07	13.26	14.50	(7)
(8)		HDD50	6159	1132	1042	888	398	88	8	1	6	90	390	897	1219	(8)
(9)		HDD65	10581	1597	1462	1353	840	447	239	139	192	435	846	1347	1684	(9)
(10)		CDD50	1090	0	0	0	8	107	224	343	294	106	8	0	0	(10)
(11)		CDD65	39	0	0	0	0	1	5	17	15	1	0	0	0	(11)
(12)		CDH74	726	0	0	0	3	54	91	276	244	58	0	0	0	(12)
(13)		CDH80	111	0	0	0	0	5	5	44	51	6	0	0	0	(13)
(14)	Wind	WSAvg	8.2	8.6	7.9	7.8	9.0	9.2	8.4	7.4	7.2	7.8	8.7	7.9	8.0	(14)
(15)	Precipitation	PrecAvg	18.1	1.2	0.9	1.0	0.8	1.3	2.9	3.1	2.0	1.3	1.1	1.2	1.2	(15)
(16)		PrecMax	25.8	2.6	1.8	2.7	1.8	3.4	6.0	5.7	4.4	3.3	3.7	3.5	5.1	(16)
(17)		PrecMin	11.3	0.4	0.0	0.2	0.2	0.3	1.0	1.1	0.3	0.3	0.3	0.2	0.2	(17)
(18)		PrecStd	3.9	0.6	0.5	0.6	0.4	0.8	1.5	1.1	1.1	0.7	0.7	0.7	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.9	43.3	49.7	68.4	80.0	80.5	84.3	85.4	79.8	66.7	47.3	42.3	(19)
(20)			MCWB	36.5	37.1	40.6	49.9	54.4	59.6	62.2	61.5	58.6	51.6	40.5	36.0	(20)
(21)		2%	DB	41.5	40.3	45.4	59.7	73.8	76.6	80.2	80.3	74.8	59.1	42.6	39.4	(21)
(22)			MCWB	36.2	35.3	37.7	45.5	52.5	57.5	60.8	60.6	56.7	47.6	37.4	34.2	(22)
(23)		5%	DB	39.6	37.5	41.9	54.0	69.5	73.7	76.9	75.9	69.4	54.3	39.9	35.6	(23)
(24)			MCWB	35.1	33.2	35.5	42.1	50.5	56.5	59.7	59.3	54.6	44.7	35.0	31.3	(24)
(25)		10%	DB	37.1	34.3	39.0	50.0	65.8	70.3	73.6	72.6	64.5	50.6	37.5	32.0	(25)
(26)			MCWB	33.1	30.7	33.7	40.0	49.0	54.9	58.5	58.0	52.4	42.3	33.3	28.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	37.8	41.4	51.0	57.2	62.8	67.0	64.9	61.0	52.9	41.7	37.1	(27)
(28)			MCD B	42.0	42.3	48.1	67.7	72.0	76.0	79.8	76.2	75.4	62.5	45.6	41.6	(28)
(29)		2%	WB	36.6	35.5	37.9	46.4	54.1	60.2	63.3	62.8	57.9	48.8	37.7	34.4	(29)
(30)			MCD B	40.9	39.8	44.4	57.6	69.0	71.9	74.1	74.6	71.2	57.2	41.7	39.2	(30)
(31)		5%	WB	35.2	33.4	35.9	42.8	52.0	58.2	61.6	61.0	55.5	45.6	35.5	31.4	(31)
(32)			MCD B	39.5	37.3	41.2	52.6	66.6	69.9	72.7	73.1	68.0	52.9	39.3	35.3	(32)
(33)		10%	WB	33.1	30.7	33.9	40.4	50.4	56.4	59.8	59.2	53.3	43.1	33.6	28.1	(33)
(34)			MCD B	37.1	34.3	38.8	49.1	63.6	67.1	70.6	70.1	63.2	49.7	37.4	31.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.7	14.7	16.3	17.3	21.7	19.7	20.5	21.3	20.1	14.5	13.7	14.0	(35)
(36)			MCDBR	13.5	14.5	17.8	23.7	27.7	25.5	27.2	27.4	27.8	21.3	15.5	15.6	(36)
(37)			MCWBR	10.5	10.9	12.0	12.9	12.5	10.9	12.1	11.9	12.8	12.3	12.0	13.1	(37)
(38)		5% WB	MCDBR	13.2	15.3	16.9	23.0	24.9	22.2	24.4	24.8	26.9	20.4	15.9	16.2	(38)
(39)			MCWBR	10.4	11.8	11.9	13.1	11.7	10.5	12.5	12.1	13.1	12.7	13.1	13.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.250	0.250	0.292	0.317	0.341	0.344	0.345	0.343	0.304	0.281	0.238	0.220	(40)
(41)		taud		2.216	2.336	2.281	2.395	2.402	2.407	2.432	2.427	2.544	2.559	2.410	2.239	(41)
(42)		Ebn at Noon		212	268	279	288	285	285	282	276	275	256	226	200	(42)
(43)		Edh at Noon		17	22	31	32	34	34	33	31	24	18	14	12	(43)
(44)	All-Sky Solar Radiation	RadAvg		222	525	1002	1492	1771	1912	1857	1512	1036	548	244	154	(44)
(45)		RadStd		23	41	76	80	121	155	111	134	88	63	21	12	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page